Part II, Lecture 6: Corporate Profitability and Bank Lending - Cash is King, but Profit is King's Father

Why this session matters

Lectures 1 to 4 measured credit risk through ratings, default probabilities and pricing. This lecture changes the angle: what the bank should look at inside the borrowing company itself. The regulator's answer (the EBA Guidelines on Loan Origination and Monitoring, EBA LOM 2020) is a 360-degree, forward-looking business analysis in which profitability is the cornerstone of creditworthiness. The practical tool is the extended additive ROE formula, which decomposes shareholder profitability into operating performance, the spread over the cost of debt, the two leverages, extraordinary items and taxes. If you can rebuild the formula step by step, apply it to two companies and explain why identical ROEs can hide very different risks, you own this lecture.

The regulatory anchor: EBA LOM 2020

> **EBA LOM 2020**: the "Final Report on the EBA Guidelines on Loan Origination and Monitoring". Core objective: ensure robust and prudent credit risk management by institutions. The big picture is not just current numbers but a 360-degree business analysis that integrates the financial and the qualitative profiles of the borrower, on a medium- to long-term horizon. Profitability is the cornerstone: it drives business sustainability and therefore borrower creditworthiness.

- After intense criticism by the financial industry of the 1999 draft, the EBA split requirements into two macro-segments of banking lending: micro and small enterprises (MS) versus medium-sized and large enterprises (MSL).
- The definitions come from Art. 2 of the Commission Recommendation of 6 May 2003:
- **Micro enterprise**: fewer than 10 persons employed, and annual turnover and/or balance sheet total not exceeding EUR 2 million.
- **Small enterprise**: fewer than 50 persons, turnover and/or balance sheet total not exceeding EUR 10 million.
- **Medium-sized enterprises (within SMEs)**: fewer than 250 persons, turnover not exceeding EUR 50 million and/or balance sheet total not exceeding EUR 43 million.
- In the final EBA LOM, many paragraphs are identical for the two segments, and even where requirements differ, the differences are minimal.

What the guidelines require banks to assess

- **Comprehensive information** (§86, §90, both segments): business model, corporate structure, business plans with financial projections.
- **Realistic projections** (§129 MS, §151 MSL): the bank must assess the borrower's projections and even build its own when borrower data is unreliable.

- **Future profitability** (§152, both segments): how retained earnings feed back into equity.
- **Stress testing** (§131 MS, §156-158 MSL): repayment capacity under adverse conditions. For MSL enterprises the EBA lists the factors to stress:
- Idiosyncratic events: decline in revenues or profit, operational loss, management problems, supplier failure, reputational damage, liquidity outflow.
- Market events: macro downturn, sector downturn, interest rate shocks (the slides cite +200 bps).
- **Early Warning Indicators (EWIs)** (§274, both segments): signals of credit quality deterioration, such as debt increase, turnover drop, narrowing margins, deviation from forecasts, rating downgrade, covenant breach.
- **Watch List** (§272): EWIs trigger an "alarm-triggered review", a deep dive similar to origination.

The lifecycle of credit assessment

- **Origination**: the initial assessment.
- **Ongoing monitoring**: behavioral (line usage, payments) and structural (EWIs).
- **Regular review**: a periodic (annual) 360-degree reassessment.
- **Watch list review**: triggered by EWIs, an "alarm" 360-degree review that may eventually change the borrower's rating.

Credit assessment is therefore not a one-off decision at loan granting: it is a continuous cycle in which monitoring can reopen the origination analysis at any time.

Asset-based lending versus cash-flow lending

Banks channel household savings to entrepreneurial initiatives; the goal is to fund the ventures that generate the highest economic value with the lowest risk, which drives GDP growth even with constant financial resources. Prioritising profitable enterprises with sustainable and predictable earnings means lower default probability. Two lending policies pursue this differently:

- **Asset-based lending** focuses on collateral (mortgages, pledges) and already strong balance sheets. It is appealing because it reduces the operational costs of credit analysis: existing assets are easier to evaluate. The problem: it does not ensure resources go to the most productive ventures, so it can hinder efficient allocation.
- **Cash-flow lending (the EBA LOM approach)** focuses on future profitability and sustainable earnings potential. It directs funds to economically viable businesses, aligning the bank with efficient resource allocation and long-term growth.

From the bank's own management perspective, cash-flow lending also wins:

- **Client portfolio erosion** under asset-based lending: defaults, even with recovery, mean losing clients, who must be replaced by potentially marginal borrowers.
- **Capital erosion**: high equity at origination does not guarantee resilience, because operational losses can quickly wipe it out.
- Cash-flow lending is forward-looking and addresses fundamental business health: it lends to firms whose viability rests on operating income.

The distress sequence: how companies get into trouble

The slides give a logical and chronological sequence:

1. Failure to perceive or interpret changes in the competitive environment.
2. Inability to innovate strategically (no new business model).
3. Failure to implement the new business model effectively.
4. **Operational losses**: core operations fail to generate adequate income, that is ROA below the cost of debt. The profitability problem emerges here.
5. Cash deficit and weakened financial structure.
6. Creditors tighten lending and distress signals emerge (Credit Registers, credit line usage).

The lesson: the profitability problem appears at step 4, before the financial structure deteriorates at steps 5 and 6. A bank that reads only balance-sheet ratios sees the trouble two steps late. Changes in the environment call for behavioral analyses (tech innovations, competitors' moves), the flow of funds and financial structure call for traditional financial analysis, and strategy, profit and organisation call for strategic analyses and modern financial analysis.

Matching risks with analysis tools

- **Strategic and operational risks**: root cause is inconsistency between environment, strategy and operations. They manifest in the operating margin of the income statement. Tools: competitive analysis, business model review, qualitative assessment, modern profitability analysis (quality of earnings assessed through a system of ratios examined together with strategic analysis).
- **Financial risk**: root cause is a weak balance sheet and mismatches in financial characteristics. It manifests in balance-sheet strength, cash flow and debt levels. Tools: financial structure analysis, traditional ratio analysis based on ideal values of individual ratios.

Early detection means identifying strategic and operational issues before they become financial problems. The slides rank the analyses on two axes: macro environment, industry and strategy analyses are the most forward-looking, while financial performance analysis (ratios, cash flow), analysis of the company's and owners' wealth, and analysis of the relationships with the banking system are temporally closest to current events.

The reclassified financial statements

Modern ratio analysis uses historical statements to identify strategic choices, management policies and external influences, under two principles: go beyond the numbers to the underlying business dynamics (the "quality" of profits), and apply a systemic view (ratios are never analysed in isolation, but as a coordinated system of indicators).

The reclassified balance sheet isolates four aggregates:

- **Total Assets (TA)**: everything the company employs.
- **Equity (E)**.
- **Financial Debt (D)**: debt that pays interest.
- **Other Liabilities (OL)**: non-interest-bearing liabilities, such as trade payables and provisions. Trade payables are treated as cost-free because the implicit cost (foregone early-payment discounts) is already accounted for in the cost of purchases.

The reclassified income statement is a ladder of subtotals: Net sales plus change in stocks of finished goods and work in progress plus capitalized costs give the VALUE OF PRODUCTION; subtract purchases of raw materials and cost for services to get ADDED VALUE; subtract staff costs and severance provisions to get GROSS OPERATING INCOME (EBITDA); subtract depreciation, write-downs on working capital and provisions to get NET OPERATING INCOME (EBIT); add or subtract other non-operating non-financial items to get CURRENT INCOME BEFORE FINANCIAL P&L; add financial income and subtract losses on financial assets to get CURRENT INCOME BEFORE INTEREST EXPENSES (CIBIE); subtract interest and other financial expenses to get CURRENT INCOME; adjust for disposals and extraordinary items to get GROSS OF TAX PROFIT; subtract taxes to reach NET PROFIT.

> **CIBIE** (Current Income Before Interest Expenses): the income generated by all assets before paying the providers of debt. It is the numerator of ROA and the pivot of the whole ROE decomposition, because it measures what the assets earn independently of how they are financed.

The extended additive ROE formula

> **Extended additive ROE formula**: ROE = [ROA + (ROA - C) x D/E + ROA x OL/E] x GP/CI x (1 - T)

Where:

- **ROE** = Net Profit / Equity: shareholder profitability.
- **ROA** = CIBIE / Total Assets: return on all assets, independent of the capital structure.
- **C** = Interest Expenses / Financial Debt: average cost of borrowing.
- **D/E**: financial leverage.
- **OL/E**: the non-interest-bearing liabilities leverage.
- **GP/CI** = Gross of tax Profit / Current Income: impact of extraordinary items.
- **T** = Taxes / Gross Profit: effective tax rate.

What the formula actually does, step by step:

- **Step 1, the simplest world** (no debt, no OL, no extraordinary items, no taxes): ROE = ROA. The numerator NP equals CIBIE and the denominator E equals TA, so owners earn exactly what the assets earn.
- **Step 2, add financial debt**: each euro of assets earns ROA; assets financed by equity pay it fully to shareholders, while assets financed by debt must first pay C. The residual spread (ROA - C) accrues to shareholders, amplified by the leverage D/E: ROE = ROA + (ROA - C) x D/E.
- **Step 3, add other liabilities**: OL pay no interest, so the ROA earned on OL-financed assets benefits shareholders in full: add ROA x OL/E.
- **Step 4, add extraordinary items and taxes**: multiply by GP/CI (above 1 when non-recurring items help, below 1 when they hurt) and by (1 - T), the share of gross earnings kept after taxes.

Reading of the components: the bracketed term is the gross ROE, equal to CI/E, profitability before extraordinary items and taxes. Inside it: ROA is the return on equity directly invested; (ROA - C) x D/E is the contribution of financial debt, the spread captured by shareholders amplified by leverage; ROA x OL/E is the contribution of cost-free liabilities. Outside: GP/CI and (1 - T) scale the result for extraordinary items and taxation. The formula is an identity equivalent to NP/E, provable by multiplying both sides by E, substituting CIBIE = CI + IE and simplifying down to NP = NP.

Same ROE, different risk

> **The slides' comparison.** Case A: ROE 17% built from ROA 20%, cost of debt 6%, spread 14%, leverage D/E of 1, tax 50%. Case B: the same ROE 17% built from ROA 14%, cost of debt 10%, spread only 4%, leverage D/E of 5, tax 50%. Do they carry the same risk for lenders? Certainly not. B's core business is far less profitable, its borrowing is more expensive, its spread is razor-thin (so the probability of ROA falling below C is higher), and its leverage is five times higher. Leverage amplifies gains when the spread is positive and amplifies losses catastrophically when it turns negative. Same ROE, completely different risk.

Alternative ROE approaches, and why the additive formula wins

A common alternative deducts OL from total assets to get Net Assets (NA) and computes ROI = CIBIE / NA, giving ROE = [ROI + (ROI - C) x D/E] x GP/CI x (1 - T). The extended additive formula is preferred for three reasons:

- **Independent OL leverage**: the additive formula shows the explicit contribution of cost-free other liabilities (ROA x OL/E); the net-assets version hides it.
- **Consistent ROA**: ROA = CIBIE/TA uses a numerator that includes all returns on all assets (operating, financial, other).
- **Clear cost of debt**: C = IE/D is computed explicitly, allowing a precise comparison of borrowing cost against asset return; netting financial income and expenses (as some alternatives do) destroys this insight.

Worked exercise: Alpha Co. versus Beta Corp.

> **Data (EUR millions).** Alpha: NP 12, E 60, CIBIE 30, TA 100, IE 4, D 40, OL 0, GP 15, CI 26, Taxes 3. Beta: NP 12, E 40, CIBIE 20, TA 100, IE 6, D 60, OL 0, GP 15, CI 14, Taxes 3.

Component ratios:

- **Alpha**: ROA = 30/100 = 30%; C = 4/40 = 10%; D/E = 40/60 = 0.667; T = 3/15 = 20%, so (1 - T) = 80%.
- **Beta**: ROA = 20/100 = 20%; C = 6/60 = 10%; D/E = 60/40 = 1.5; T = 20%, so (1 - T) = 80%.

Applying the formula (OL = 0 for both, so the OL term vanishes):

- **Alpha**: ROE = [0.30 + (0.30 - 0.10) x 0.667] x (15/26) x 0.80 = 20%, which matches NP/E = 12/60.
- **Beta**: ROE = [0.20 + (0.20 - 0.10) x 1.5] x (15/14) x 0.80 = 30%, which matches NP/E = 12/40.

Which company is riskier for a lender? Beta's ROE (about 30%) is 50% higher than Alpha's (about 20%), largely because of its much higher financial leverage. But comparing gross ROE (CI/E) reverses the picture: Alpha earns 43% (26/60) against Beta's 35% (14/40). Alpha has the stronger underlying operating performance; its lower reported ROE was caused by a large non-recurring negative extraordinary item, which needs investigation, while Beta's reported ROE was boosted by a small non-recurring positive item. The decomposition, not the headline ROE, tells the risk story.

What you must walk away with

Must know:

• EBA LOM 2020: 360-degree, forward-looking analysis; profitability is the cornerstone of creditworthiness; separate MS and MSL segments with minimal differences.

• Required assessments: comprehensive information, realistic projections, retained earnings, stress testing (idiosyncratic and market events, e.g. +200 bps), EWIs (debt increase, turnover drop, narrowing margins, forecast deviation, downgrade, covenant breach), watch list with alarm-triggered review.

• Credit assessment is a lifecycle: origination, ongoing behavioral and structural monitoring, annual review, alarm review that can change the rating.

• Cash-flow lending beats asset-based lending: it funds future viability, avoids client portfolio erosion and capital erosion.

• Distress sequence: missed environmental change, no strategic innovation, failed implementation, operational losses (ROA below cost of debt), cash deficit, credit tightening. Profitability breaks before the balance sheet does.

• Reclassified statements: TA = E + D + OL; the income ladder runs through EBITDA, EBIT and CIBIE down to net profit.

• ROE = [ROA + (ROA - C) x D/E + ROA x OL/E] x GP/CI x (1 - T), an identity with NP/E built up in four steps.

• Same ROE can hide different risk: read the spread (ROA - C), the leverage and the gross ROE (CI/E), never the headline number alone.

• Alpha versus Beta: Beta's higher ROE comes from leverage, not performance; Alpha's gross ROE of 43% versus 35% reveals the stronger operator.